US006816325B1

(12) **United States Patent**
Becker et al.

(10) **Patent No.: US 6,816,325 B1**
(45) **Date of Patent: Nov. 9, 2004**

(54) **MOUNTING APPARATUS FOR AN OPTICAL ELEMENT**

(75) Inventors: **Jochen Becker**, Rengsdorf (DE); **Bernhard Gellrich**, Aalen (DE); **Werner Lang**, Geislingen (DE); **Alexander Kohl**, Aalen (DE); **Sascha Kraus**, Aalen (DE)

(73) Assignee: **Carl Zeiss SMT AG**, Oberkochen (DE)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **10/661,183**

(22) Filed: **Sep. 11, 2003**

(51) **Int. Cl.**[7] **G02B 7/02**

(52) **U.S. Cl.** **359/819**; 359/822; 359/811; 359/813

(58) **Field of Search** 359/819, 822, 359/823, 826, 811, 813

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,917,385	A	11/1975	Caswell	359/822
5,428,482	A	6/1995	Bruning et al.	359/827
5,537,262	A	7/1996	Aoki et al.	359/822
6,144,511	A *	11/2000	Umetsu et al.	359/831
6,275,344	B1	8/2001	Holderer	359/822
6,307,688	B1	10/2001	Merz et al.	359/819
6,580,570	B2	6/2003	Becker et al.	359/819
6,603,615	B2	8/2003	Melzer et al.	359/823
2002/0001142	A1	1/2002	Osterried	359/819
2002/0021504	A1	2/2002	Bayer et al.	359/827

FOREIGN PATENT DOCUMENTS

CH	371906	10/1963
DE	1 675 685	1/1970
DE	30 30 549 A1	2/1982
DE	3406907 A 1	10/1984
DE	41 36 580 A1	5/1993
DE	199 08 554.4	3/1999
DE	198 25 716 A 1	12/1999
DE	199 10 947 A 1	9/2000
EP	0 053 463 A2	6/1982
EP	0 230 277 A2	7/1987
EP	0 597 209 A1	9/1993
EP	0 964 281 A1	12/1999
EP	1 209 500 A2	5/2002

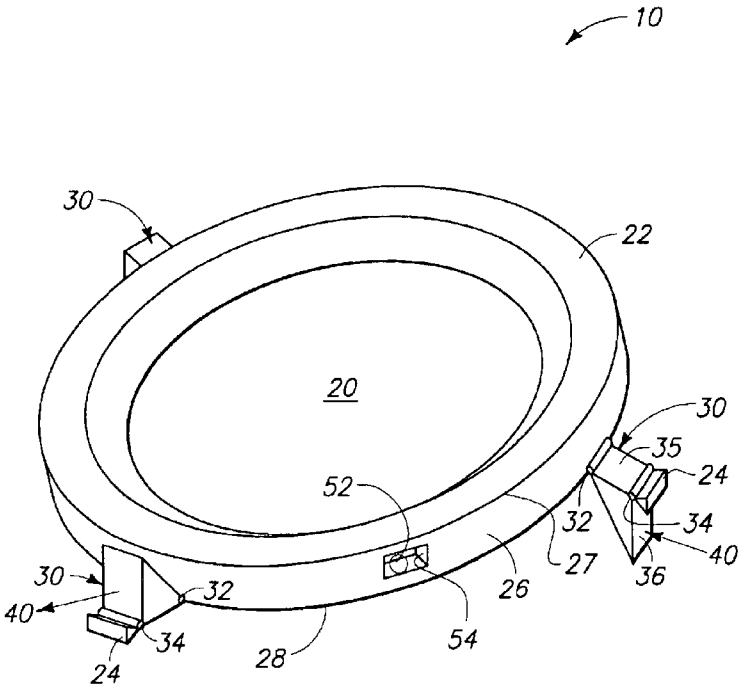
* cited by examiner

Primary Examiner—Tim Thompson
(74) *Attorney, Agent, or Firm*—Wells St. John P.S.

(57) **ABSTRACT**

In the case of a mounting apparatus, an optical element having an inner mount and an outer mount, in particular a lens in a projection lens system for semiconductor lithography, the inner mount is connected to the outer mount via three circumferentially distributed articulations. Manipulators, whereby said inner mount is displaceable, act on the articulations. The articulations comprise a mechanism which transforms a radial movement into an axial movement.

45 Claims, 2 Drawing Sheets



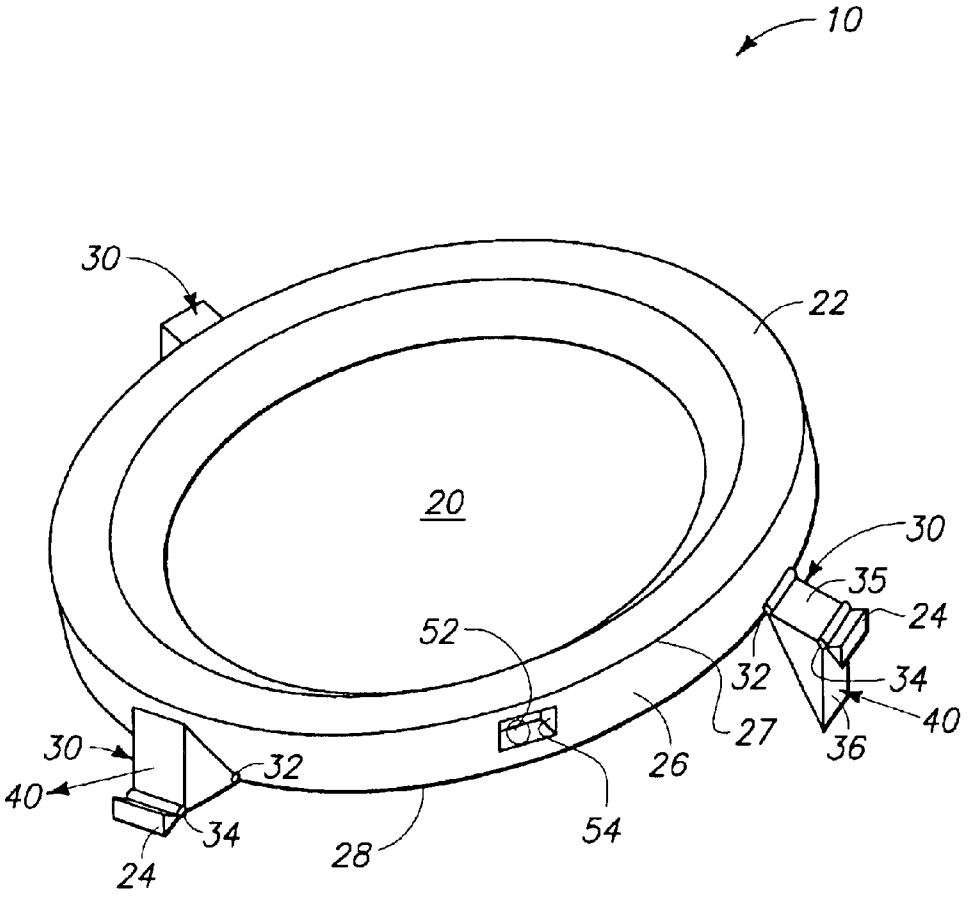


FIG. 1

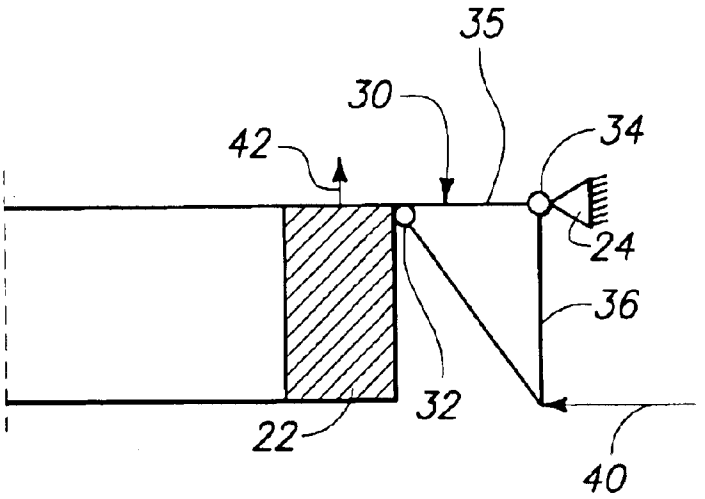


Fig 2

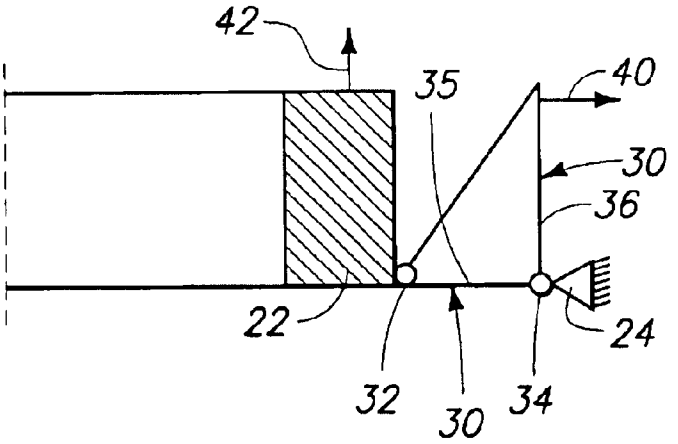


Fig 3

MOUNTING APPARATUS FOR AN OPTICAL ELEMENT

TECHNICAL FIELD

The invention relates to a mounting apparatus for an optical element and more particularly to a mounting apparatus for mounting a lens.

BACKGROUND OF THE INVENTION

Optical elements, e.g. lenses, have to be installed and adjusted very precisely in relation to their mechanical reference, in particular, in lens systems for semiconductor lithography. It is thus the case, eg. with lenses, that the optical axis has to be made to coincide as precisely as possible with the ideal mechanical axis.

The Applicant's earlier application P 199 08 554.4 discloses a three-point mounting via solid-state rotary articulations with solid-state transitions which can be adjusted by manipulators. A three-point mounting is also known from U.S. Pat. No. 3,917,385.

SUMMARY OF THE INVENTION

The object of the present invention is to improve the hitherto known installation and adjustment mounting method by an integrated fine-tuning functional unit, in order to achieve higher positioning accuracy.

Methods and apparatuses for achieving this object are described. In one embodiment, a mounting apparatus for an optical element includes an inner mount and an outer mount, the inner mount being connected to the outer mount via three circumferentially distributed articulations. An exemplary mounting apparatus is disclosed in U.S. Pat. No. 6,580,570, issued Jun. 17, 2003, to Becker et al., which is incorporated herein by reference. Manipulators act on the articulations wherein said inner mount is displaceable, and wherein the articulations comprise a mechanism which transforms a radial movement into an axial movement. In another embodiment, the mounting apparatus is for mounting a lens in a projection lens system for semiconductor lithography. In another embodiment, the articulations comprise at least one lever for transforming the radial movement into an axial movement, wherein the manipulators act on said at least one lever. In another embodiment, the articulations comprise at least one bell crank lever with attachment points between the inner mount and the outer mount. The outer mount is located in the region of the outer ends of a first lever of the at least one bell crank lever and the manipulators act in the region of the outer end of a second lever of the at least one bell crank lever which is in an angle to the first lever.

In another embodiment, the articulations comprise at least one T-shaped lever in cross section, with a T-bar and a T-support, attachment points between the inner mount and the outer mount being located in each case in the region of the outer ends of the T-bar, and the manipulators acting on the T-support in each case, this embodiment being disclosed in U.S. Pat. No. 6,580,570, previously incorporated herein by reference.

BRIEF DESCRIPTION OF THE DRAWINGS

Preferred embodiments of the invention are described below with reference to the following accompanying drawings.

FIG. **1** shows a basic perspective illustration of the three-point mounting with the articulations according to the invention;

FIG. **2** shows a basic illustration of a first embodiment of an articulation; and

FIG. **3** shows a basic illustration of a second embodiment of an articulation.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

This disclosure of the invention is submitted in furtherance of the constitutional purposes of the U.S. Patent Laws "to promote the progress of science and useful arts" (Article 1, Section 8).

According to the illustration in FIG. **1**, an exemplary mounting apparatus **10** is shown. An optical element **20** is mounted in an inner mount **22**, which is connected to an outer mount **24** via a plurality of circumferentially distributed articulations **30**. An exemplary optical element includes a lens **20** for a projection lens system sued in semiconductor lithography processing. The number of exemplary plurality articulations **30** include three as shown. However, it should be understood that one or any number of articulations **30** could be used in the scope of this invention, and oriented circumferentially in any matter along inner mount **22**. For reasons of clarity, the outer mount **24** is only indicated at the attachment points in FIG. **1**. Inner mount **22** has a circular sidewall **26** with an upper edge **27** and lower edge **28**. An exemplary articulation **30** includes a bell crank lever **30** rotatably secured between inner and outer mounts **22** and **24** by pivots or attachment points **32** and **34**. Attachment points **32** and **34** are located between the inner mount **22** and the outer mount **24** and at an outer end of a first lever **35** of the bell crank lever **30**. A second lever **36** of the bell crank lever **30** is located perpendicularly or rectangularly to the first lever **35**. A manipulator **40** (referenced only by direction arrow **40**) provides a force to the outer end of the second lever **36** and subjects second lever **36** to tensile and/or compressive forces. As a result of the force by manipulator **40** and the kinematics of the bell crank lever **30**, the inner mount **22** is displaced in an upward or a downward direction by transforming a radial movement into an axial movement. Exemplary manipulators include mechanical, electrical, hydraulic and pneumatic actuating members as understood by those skilled in the art, and an exemplary location for such manipulator is in the outer mount **24** (not shown).

FIG. **1** further illustrates in principle how position determination can be carried out, for example, by a capacitive sensor **52**, which may be located in a recess **54** between the inner mount **22** and the outer mount **24**. For this purpose, of course, it is necessary to have a plurality of capacitive sensors **52** correspondingly distributed over the circumference of inner mount **22**.

FIG. **2** shows a first exemplary embodiment of an articulation illustrated as a bell crank lever **30**. The attachment points **32** and **34** are positioned between the inner mount **22** and the outer mount **24** generally coextensive with the upper edge **27** of circular sidewall **26** of inner mount **22**. In this orientation of bell crank lever **30**, the second lever **36** extends downward from attachment point **34** generally spaced from and along circular sidewall **26**. Manipulator **40** provides a compressive force to the outer end of the second lever **36** and moves the outer end of the second lever **36** radially to drive inner mount **22** and optical element **20** axially upward in direction **42**. Alternatively, manipulator **40** provides a tensile force to the outer end of the second lever **36** and moves the outer end of the second lever **36** radially to drive inner mount **22** and optical element **20** axially in a direction opposite to direction **42**.

FIG. **3** shows a second exemplary embodiment of an articulation illustrated as a bell crank lever **30**. The attachment points **32** and **34** are positioned between the inner mount **22** and the outer mount **24** generally coextensive with the lower edge **27** of circular sidewall **26** of inner mount **22**. In this orientation of bell crank lever **30**, the second lever **36** extends upward from attachment point **34** generally spaced from and along circular sidewall **26**. Manipulator **40** provides a tensile force to the outer end of the second lever **36** and moves the outer end of the second lever **36** radially to drive inner mount **22** and optical element **20** axially upward in direction **42**. Alternatively, manipulator **40** provides a compressive force to the outer end of the second lever **36** and moves the outer end of the second lever **36** radially to drive inner mount **22** and optical element **20** axially in a direction opposite to direction **42**.

It should be understood that manipulator **30** can be rotatably secured to inner mount **22** by attachment point **34** between upper and lower edges **27** and **28** at any location of circular sidewall **26**. It should be understood that any combination of articulations **30** can be used, for example, all the articulations **30** secured to mounting apparatus **10** could be bell crank levers, or all could be the one T-shaped articulations disclosed in U.S. Pat. No. 6,580,570, previously incorporated herein by reference, or any combination of differently designed articulations.

In compliance with the statute, the invention has been described in language more or less specific as to structural and methodical features. It is to be understood, however, that the invention is not limited to the specific features shown and described, since the means herein disclosed comprise preferred forms of putting the invention into effect. The invention is, therefore, claimed in any of its forms or modifications within the proper scope of the appended claims appropriately interpreted in accordance with the doctrine of equivalents.

The invention claimed is:

1. A mounting apparatus for an optical element, comprising:

an inner mount;

an outer mount, the inner mount being connected to the outer mount via three circumferentially distributed articulations; and

manipulators configured to act on the articulations, wherein said inner mount is displaceable, and wherein the articulations comprise a mechanism which transforms a radial movement into an axial movement.

2. The apparatus as claimed in claim **1**, wherein said mechanism comprises at least one lever for transforming the radial movement into an axial movement, wherein said manipulators act on said at least one lever.

3. The apparatus as claimed in claim **2**, wherein said mechanism comprises at least one bell crank lever with attachment points between said inner mount and said outer mount and being located in the region of the outer ends of a first lever of said at least one crank lever, and said manipulators acting in the region of the outer end of a second lever of said at least one crank lever which is in an angle to said first lever of said at least one crank lever.

4. The apparatus as claimed in claim **3**, wherein said first and said second lever of said at least one crank lever are perpendicular to each other.

5. The apparatus as claimed in claim **2**, wherein said lever of said mechanism changes the transmission proportion of said transfer from radial movement to axial movement.

6. The apparatus as claimed in claim **5**, wherein said lever of said mechanism is located outside of the geometry of said inner mount and of said outer mount resulting in a greater force to said inner mount.

7. The apparatus as claimed in claim **1**, wherein said articulations are built up from separate parts.

8. The apparatus as claimed in claim **1**, further comprising sensors provided for position determination of said inner mount.

9. The apparatus as claimed in claim **8**, wherein said sensors are designed as capacitive sensors.

10. The apparatus as claimed in claim **1**, wherein said manipulators provided are hydraulic or pneumatic actuating members.

11. The apparatus as claimed in claim **1**, wherein said manipulators provided are mechanical actuating members.

12. The apparatus as claimed in claim **1**, wherein said manipulators provided are electrical actuating members.

13. The apparatus as claimed in claim **1**, wherein said manipulators for the individual articulations can be actuated separately in each case.

14. A mounting apparatus for mounting a lens in a projection lens system for semiconductor lithography, comprising:

an inner mount;

an outer mount, the inner mount being connected to the outer mount via three circumferentially distributed articulations; and

manipulators configured to act on the articulations, wherein said inner mount is displaceable, and wherein the articulations comprise a mechanism which transforms a radial movement into an axial movement.

15. The apparatus as claimed in claim **14**, wherein said mechanism comprises at least one lever for transforming the radial movement into an axial movement, wherein said manipulators act on said at least one lever.

16. The apparatus as claimed in claim **15**, wherein said mechanism comprises at least one bell crank lever with attachment points between said inner mount and said outer mount being located in the region of the outer ends of a first lever of said at least one crank lever, and said manipulators acting in the region of the outer end of a second lever of said at least one crank lever which is in an angle to said first lever of said at least one crank lever.

17. The apparatus as claimed in claim **16**, wherein said first and said second lever of said at least one crank lever are perpendicular to each other.

18. The apparatus as claimed in claim **14**, wherein said manipulators for said individual articulations can be actuated separately in each case.

19. The apparatus as claimed in claim **14**, further comprising sensors provided for position determination of the inner mount.

20. The apparatus as claimed in claim **19**, wherein said sensors are designed as capacitive sensors.

21. The apparatus as claimed in claim **14**, wherein said manipulators provided are hydraulic or pneumatic actuating members.

22. The apparatus as claimed in claim **14**, wherein said manipulators provided are mechanical actuating members.

23. The apparatus as claimed in claim **14**, wherein said manipulators provided are electrical actuating members.

24. A mounting apparatus for an optical element, comprising:

an outer mount;

an inner mount arranged within said outer mount;

an optical element being adapted to said inner mount; and

at least three articulations circumferentially arranged around said inner mount and between said inner mount

and said outer mount, wherein each of said at least three articulations comprise a mechanism which transforms a radial movement into an axial movement.

25. The mounting apparatus of claim **24**, wherein said manipulators provide a direction of force which is perpendicular to said axial movement.

26. The apparatus as claimed in claim **24**, wherein said mechanism comprises at least one lever for transforming the radial movement into the axial movement, wherein said manipulators act on said at least one lever.

27. The apparatus as claimed in claim **24**, wherein said mechanism comprises at least one bell crank lever with attachment points between said inner mount and said outer mount being located in the region of the outer ends of a first lever of said at least one crank lever, and said manipulators acting in the region of the outer end of a second lever of said at least one crank lever which is in an angle to said first lever of said at least one crank lever.

28. The apparatus as claimed in claim **27**, wherein said first and said second lever of said at least one crank lever are perpendicular to each other.

29. The apparatus as claimed in claim **24**, further comprising sensors provided for position determination of the inner mount.

30. The apparatus as claimed in claim **29**, wherein said sensors are designed as capacitive sensors.

31. The apparatus as claimed in claim **24**, wherein said manipulators provided are hydraulic or pneumatic actuating members.

32. The apparatus as claimed in claim **24**, wherein said manipulators provided are mechanical actuating members.

33. The apparatus as claimed in claim **24**, wherein said manipulators provided are electrical actuating members.

34. A mounting apparatus for mounting a lens in a projection lens system for semiconductor lithography, comprising:

an outer mount;

an inner mount arranged within said outer mount;

an optical element being adapted to said inner mount; and

at least three articulations circumferentially arranged around said inner mount and between said inner mount and said outer mount, wherein each of said at least three articulations comprise a mechanism which transforms a radial movement into an axial movement.

35. The mounting apparatus of claim **34**, wherein said manipulators providing a direction of force which is perpendicular to said axial movement.

36. The apparatus as claimed in claim **34**, wherein said mechanism comprises at least one lever for transforming the radial movement into an axial movement, wherein said manipulators act on said at least one lever.

37. The apparatus as claimed in claim **36**, wherein said mechanism comprises at least one bell crank lever with attachment points between said inner mount and said outer mount being located in the region of the outer ends of a first lever of said at least one crank lever, and said manipulators acting in the region of the outer end of a second lever of said at least one crank lever which is in an angle to said first lever of said at least one crank lever.

38. The apparatus as claimed in claim **34**, wherein said first and said second lever of said at least one crank lever are perpendicular to each other.

39. The apparatus as claimed in claim **34**, further comprising sensors provided for position determination of the inner mount.

40. The apparatus as claimed in claim **39**, wherein the sensors are designed as capacitive sensors.

41. The apparatus as claimed in claim **34**, wherein the manipulators provided are hydraulic or pneumatic actuating members.

42. The apparatus as claimed in claim **34**, wherein the manipulators provided are mechanical actuating members.

43. The apparatus as claimed in claim **34**, wherein the manipulators provided are electrical actuating members.

44. The apparatus as claimed in claim **36**, wherein said lever of said mechanism changes the transmission proportion of said transfer from radial movement to axial movement.

45. The apparatus as claimed in claim **44**, wherein said lever of said mechanism is located outside of the geometry of said inner mount and of said outer mount resulting in a greater force to said inner mount.

* * * * *